Rootkit

Adversaries may use rootkits to hide the presence of programs, files, network connections, services, drivers, and other system components. Rootkits are programs that hide the existence of malware by intercepting/hooking and modifying operating system API calls that supply system information. [1]

Rootkits or rootkit enabling functionality may reside at the user or kernel level in the operating system or lower, to include a hypervisor or System Firmware. [2] Rootkits have been seen for Windows, Linux, and Mac OS X systems. [3] [4]

Rootkits that reside or modify boot sectors are known as Bootkits and specifically target the boot process of the operating system.

ID: T1014

Sub-techniques: No sub-techniques

ⓘ Tactic: Stealth

ⓘ Platforms: Linux, Windows, macOS

Contributors: Menachem Goldstein

Version: 2.0

Created: 31 May 2017

Last Modified: 12 May 2026

Version Permalink

Procedure Examples

ID	Name	Description
G0007	APT28	APT28 has used a UEFI (Unified Extensible Firmware Interface) rootkit known as LoJax.[5][6]
G0096	APT41	APT41 deployed rootkits on Linux systems.[7][8]
C0046	ArcaneDoor	ArcaneDoor included hooking the `processHostScanReply()` function on victim Cisco ASA devices.[9]
S0484	Carberp	Carberp has used user mode rootkit techniques to remain hidden on the system.[10]
S0572	Caterpillar WebShell	Caterpillar WebShell has a module to use a rootkit on a system.[11]
S1105	COATHANGER	COATHANGER hooks or replaces multiple legitimate processes and other functions on victim devices.[12]
S0502	Drovorub	Drovorub has used a kernel module rootkit to hide processes, files, executables, and network artifacts from user space view.[13]
S0377	Ebury	Ebury acts as a user land rootkit using the SSH service.[14][15]
S0047	Hacking Team UEFI Rootkit	Hacking Team UEFI Rootkit is a UEFI BIOS rootkit developed by the company Hacking Team to persist remote access software on some targeted systems.[16]
S0394	HiddenWasp	HiddenWasp uses a rootkit to hook and implement functions on the system.[17]
S0135	HIDEDRV	HIDEDRV is a rootkit that hides certain operating system artifacts.[18]
S0009	Hikit	Hikit is a Rootkit that has been used by Axiom.[19] [20]
S0601	Hildegard	Hildegard has modified /etc/ld.so.preload to overwrite readdir() and readdir64().[21]
S0040	HTRAN	HTRAN can install a rootkit to hide network connections from the host OS.[22]
S1186	Line Dancer	Line Dancer can hook both the crash dump process and the Autehntication, Authorization, and Accounting (AAA) functions on compromised machines to evade forensic analysis and authentication mechanisms.[9]
S0397	LoJax	LoJax is a UEFI BIOS rootkit deployed to persist remote access software on some targeted systems.[6]
S1220	MEDUSA	MEDUSA is a rootkit with command execution and credential logging capabilities.[23]
S0012	PoisonIvy	PoisonIvy starts a rootkit from a malicious file dropped to disk.[24]
S0458	Ramsay	Ramsay has included a rootkit to evade defenses.[25]
C0056	RedPenguin	During RedPenguin, UNC3886 used rootkits such as REPTILE and MEDUSA.[26]
S1219	REPTILE	REPTILE has the ability to hook kernel functions and modify functions data to achieve rootkit functionality such as hiding processes and network connections.[23]
G0106	Rocke	Rocke has modified /etc/ld.so.preload to hook libc functions in order to hide the installed dropper and mining software in process lists.[27]
S0468	Skidmap	Skidmap is a kernel-mode rootkit that has the ability to hook system calls to hide specific files and fake network and CPU-related statistics to make the CPU load of the infected machine always appear low.[28]

ID	Name	Description
S0221	Umbreon	Umbreon hides from defenders by hooking libc function calls, hiding artifacts that would reveal its presence, such as the user account it creates to provide access and undermining strace, a tool often used to identify malware.[32]
G1048	UNC3886	UNC3886 has used the publicly available rootkits REPTILE and MEDUSA on targeted VMs.[23]
S0022	Uroburos	Uroburos can use its kernel module to prevent its host components from being listed by the targeted system's OS and to mediate requests between user mode and concealed components.[33][34]
S0670	WarzoneRAT	WarzoneRAT can include a rootkit to hide processes, files, and startup.[35]
S0430	Winnti for Linux	Winnti for Linux has used a modified copy of the open-source userland rootkit Azazel, named libxselinux.so, to hide the malware's operations and network activity.[36]
G0044	Winnti Group	Winnti Group used a rootkit to modify typical server functionality.[37]
S0027	Zeroaccess	Zeroaccess is a kernel-mode rootkit.[38]

Mitigations

This type of attack technique cannot be easily mitigated with preventive controls since it is based on the abuse of system features.

Detection Strategy

ID	Name	Analytic ID	Analytic Description
DET0377	Detection of Kernel/User-Level Rootkit Behavior Across Platforms	AN1061	Unauthorized or anomalous loading of kernel-mode drivers or DLLs, concealed services, or abnormal modification of boot components indicative of rootkit activity.
		AN1062	Abnormal loading of kernel modules, direct tampering with /dev, /proc, or LD_PRELOAD behaviors hiding processes or files.
		AN1063	Execution of unsigned kernel extensions (KEXTs), tampering with LaunchDaemons, or userspace hooks into system libraries.

References

1. Symantec. (n.d.). Windows Rootkit Overview. Retrieved December 21, 2017.
2. Wikipedia. (2016, June 1). Rootkit. Retrieved June 2, 2016.
3. Kurtz, G. (2012, November 19). HTTP iframe Injecting Linux Rootkit. Retrieved December 21, 2017.
4. Pan, M., Tsai, S. (2014). You can't see me: A Mac OS X Rootkit uses the tricks you haven't known yet. Retrieved December 21, 2017.
5. Symantec Security Response. (2018, October 04). APT28: New Espionage Operations Target Military and Government Organizations. Retrieved November 14, 2018.
6. ESET. (2018, September). LOJAX First UEFI rootkit found in the wild, courtesy of the Sednit group. Retrieved July 2, 2019.
7. Fraser, N., et al. (2019, August 7). Double DragonAPT41, a dual espionage and cyber crime operation APT41. Retrieved September 23, 2019.
8. Crowdstrike. (2020, March 2). 2020 Global Threat Report. Retrieved December 11, 2020.
9. Cisco Talos. (2024, April 24). ArcaneDoor - New espionage-focused campaign found targeting perimeter network devices. Retrieved January 6, 2025.
10. Giuliani, M., Allievi, A. (2011, February 28). Carberp - a modular information stealing trojan. Retrieved September 12, 2024.
11. ClearSky Cyber Security. (2021, January). “Lebanese Cedar” APT Global Lebanese Espionage Campaign Leveraging Web Servers. Retrieved February 10, 2021.
12. Dutch Military Intelligence and Security Service (MIVD) & Dutch General Intelligence and Security Service (AIVD). (2024, February 6). Ministry of Defense of the Netherlands uncovers COATHANGER, a stealthy Chinese FortiGate RAT. Retrieved February 7, 2024.
13. NSA/FBI. (2020, August). Russian GRU 85th GTsSS Deploys Previously Undisclosed Drovorub Malware. Retrieved August 25, 2020.
14. Vachon, F. (2017, October 30). Windigo Still not Windigone: An Ebury Update . Retrieved February 10, 2021.
15. Marc-Etienne M.Léveillé. (2024, May 1). Ebury is alive but unseen. Retrieved May 21, 2024.
16. Lin, P. (2015, July 13). Hacking Team Uses UEFI BIOS Rootkit to Keep RCS 9 Agent in Target Systems. Retrieved December 11, 2015.
17. Sanmillan, I. (2019, May 29). HiddenWasp Malware Stings Targeted Linux Systems. Retrieved June 24, 2019.
18. ESET. (2016, October). En Route with Sednit - Part 3: A Mysterious Downloader. Retrieved November 21, 2016.
19. Glyer, C., Kazanciyan, R. (2012, August 20). The “Hikit” Rootkit: Advanced and Persistent Attack Techniques (Part 1). Retrieved November 17, 2024.
20. Glyer, C., Kazanciyan, R. (2012, August 22). The “Hikit” Rootkit: Advanced and Persistent Attack Techniques (Part 2). Retrieved November 17, 2024.
21. Chen, J. et al. (2021, February 3). Hildegard: New TeamTNT Cryptojacking Malware Targeting Kubernetes. Retrieved April 5, 2021.
22. The Australian Cyber Security Centre (ACSC), the Canadian Centre for Cyber Security (CCCS), the New Zealand National Cyber Security Centre (NZ NCSC), CERT New Zealand, the UK National Cyber Security Centre (UK NCSC) and the US National Cybersecurity and Communications Integration Center (NCCIC). (2018, October 11). Joint report on publicly available hacking tools. Retrieved March 11, 2019.
23. Punsaen Boonyakarn, Shawn Chew, Logeswaran Nadarajan, Mathew Potaczek, Jakub Jozwiak, and Alex Marvi. (2024, June 18). Cloaked and Covert: Uncovering UNC3886 Espionage Operations. Retrieved September 24, 2024.
24. Hayashi, K. (2005, August 18). Backdoor.Darkmoon. Retrieved February 23, 2018.
25. Sanmillan, I.. (2020, May 13). Ramsay: A cyber-espionage toolkit tailored for air-gapped networks. Retrieved May 27, 2020.
26. Lamparski, L. et al. (2025, March 11). Ghost in the Router: China-Nexus Espionage Actor UNC3886 Targets Juniper Routers. Retrieved June 24, 2025.
27. Anomali Labs. (2019, March 15). Rocke Evolves Its Arsenal With a New Malware Family Written in Golang. Retrieved April 24, 2019.
28. Remillano, A., Urbanec, J. (2019, September 19). Skidmap Linux Malware Uses Rootkit Capabilities to Hide Cryptocurrency-Mining Payload. Retrieved June 4, 2020.
29. Nicolas Falliere, Liam O Murchu, Eric Chien 2011, February W32.Stuxnet Dossier (Version 1.4) Retrieved November 17, 2024.
30. Fiser, D. Oliveira, A. (n.d.). Tracking the Activities of TeamTNT A Closer Look at a Cloud-Focused Malicious Actor Group. Retrieved September 22, 2021.
31. Darin Smith. (2022, April 21). TeamTNT targeting AWS, Alibaba. Retrieved August 4, 2022.
32. Fernando Mercês. (2016, September 5). Pokémon-themed Umbreon Linux Rootkit Hits x86, ARM Systems. Retrieved March 5, 2018.
33. Kaspersky Lab's Global Research and Analysis Team. (2014, August 7). The Epic Turla Operation: Solving some of the mysteries of Snake/Uroburos. Retrieved December 11, 2014.
34. FBI et al. (2023, May 9). Hunting Russian Intelligence “Snake” Malware. Retrieved June 8, 2023.
35. Harakhavik, Y. (2020, February 3). Warzone: Behind the enemy lines. Retrieved December 17, 2021.
36. Chronicle Blog. (2019, May 15). Winnti: More than just Windows and Gates. Retrieved April 29, 2020.
37. Kaspersky Lab's Global Research and Analysis Team. (2013, April 11). Winnti. More than just a game. Retrieved February 8, 2017.
38. Wyke, J. (2012, April). ZeroAccess. Retrieved July 18, 2016.